One Step Function Explains Why Galaxies Die

Gas supply versus gas depletion sets a 1-Gyr timescale that unifies black hole and halo quenching.

arXiv:2610.20505 — Quenching as a step-function response from gas mass conservation

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Background

Galaxies stop forming stars through many routes, from black hole feedback to halo gas heating. A single framework linking these mechanisms has remained elusive.

> **No unified, physically motivated description connects diverse quenching processes to observable quenched fractions across galaxy populations.**

What this paper is selling

Deriving a step-function quenching response from gas mass conservation lets the authors infer gas-supply timescales directly from quenched fractions. Applied to 350,000 galaxies, it separates black hole and halo effects.

Key results to remember

- Characteristic gas-supply timescale is about 1 Gyr across the sample.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface density marks a critical state for quenching and rejuvenation.

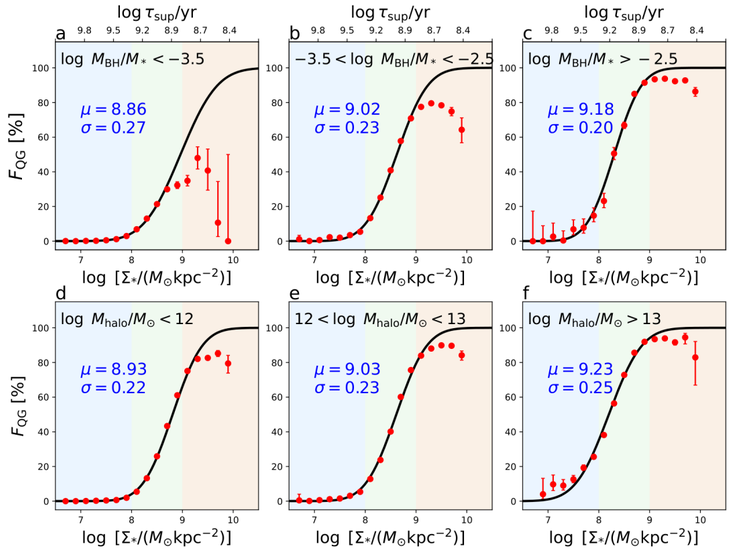


Figure 2 | Black hole growth lengthens and narrows supply timescales, while halo mass lengthens and broadens them, separating their quenching roles.

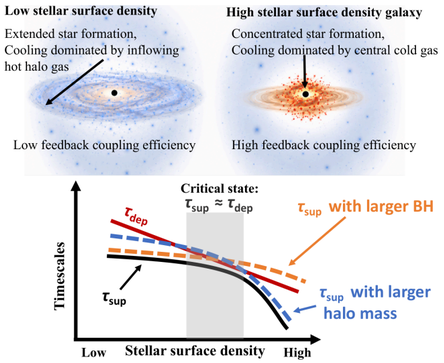


Figure 4 | A schematic picture: halo cooling governs extended star formation, while black hole feedback raises supply timescales in centrally concentrated galaxies.